Yannick Markhof

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Google Scholar Profile
Website
Latest version of this CV

Fields: Development Economics, Public Economics, Applied Econometrics & Measurement

Positions

- **ETH Zurich**, Zurich, Switzerland — 10/2025–present
  Postdoctoral Researcher, Development Economics Group; Associated Postdoc, ETH AI Center
  - Pre-job-market appointment
- **The World Bank**, Rome, Italy — 11/2020–09/2025
  Consultant, Development Data Group (LSMS) and Development Research Group
- **United Nations Development Programme**, Brasília, Brazil — 09/2019–10/2020
  Carlo-Schmid Fellow (DAAD) and Consultant

Education

- **United Nations University (UNU-MERIT)**, Maastricht, Netherlands — 2021–01/2027
  PhD in Economics (degree awarded by Maastricht University)
  - Dissertation: *Data Gaps and Economic Development.* Fully funded PhD fellowship.
- **University of California, Berkeley**, USA — 2024
  Visiting PhD Student, Department of Economics
  - Host: Edward Miguel
- **University of Oxford**, United Kingdom — 2018–2019
  MSc Economics for Development — Distinction
  - Luca D'Agliano Prize for the best MSc dissertation
- **University of Tübingen**, Germany — 2014–2018
  BSc International Economics — Grade: 1.4 (highest honors)
  - Deutschlandstipendium scholarship for outstanding academic achievement
- **University of North Carolina at Chapel Hill**, USA — 2016
  Exchange semester — GPA: 4.0, Dean's List

References

Eleonora Nillesen
UNU-MERIT / Maastricht University
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Edward Miguel
University of California, Berkeley
Dissertation committee member
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Jonathan de Quidt
Queen Mary University of London
Dissertation committee member
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Job Market Paper

Inference under measurement uncertainty: Evidence from surveys of tax compliance
with Stephan Dietrich and Rose Vincent [Latest Draft]

Abstract. Empirical research neglects the role of measurement uncertainty—uncertainty arising from discretionary measurement choices between plausible alternatives—when drawing inferences. We show that this can result in fundamentally flawed inference and distort our understanding of economic behavior. In surveys with 3,400 business owners about their tax compliance, validated against administrative tax records, measurement uncertainty dominates sampling uncertainty. In a series of experiments and tests, we trace measurement uncertainty to easily missed biases from cognitive complexity, framing, and enumerator heterogeneity. We develop a Bayesian hierarchical framework that incorporates measurement uncertainty into estimation and offers full-information inference in many applied data settings. When researchers make discretionary measurement choices, conventional standard errors understate the uncertainty around an estimate. This changes how we should draw inferences, what pre-registration can deliver, and the returns to better measurement.

Publications

Beyond the records: Data quality and COVID-19 vaccination progress in low- and middle-income countries. ***Journal of Development Economics***, 2025 (with P. Wollburg and A. Zezza). [Link]

Smallholder Farming, Fertilizer Use, and the Polycrisis Period: Cross-Country Evidence from Longitudinal Surveys in Sub-Saharan Africa. ***Food Policy***, 2025 (with A. Amankwa, A. Ambel, S. Gourlay, T. Kilic, and P. Wollburg). [Link]

Substantial impacts of climate shocks in African smallholder agriculture. ***Nature Sustainability***, 2024 (with P. Wollburg, T. Bentze, and G. Ponzini). [Link]

Assessing COVID-19 vaccine hesitancy, barriers, and uptake in Sub-Saharan Africa. ***Communications Medicine***, 2023 (with P. Wollburg, S. Kanyanda, and A. Zezza). [Link]

The Evolution of COVID-19 Vaccine Hesitancy in Sub-Saharan Africa. ***BMC Proceedings***, 2023 (with P. Wollburg, S. Kanyanda, and A. Zezza). [Link]

Acceptance of COVID-19 vaccines in Sub-Saharan Africa: Evidence from six national phone surveys. ***BMJ Open***, 2021 (with P. Wollburg, S. Kanyanda, and A. Zezza). [Link]

Working Papers and Work in Progress

Applied econometrics and measurement

- **Inference under measurement uncertainty: Evidence from surveys of tax compliance** *(Job Market Paper)*
  with S. Dietrich and R. Vincent [Latest Draft]
- **The effect of survey mode on data quality: Experimental evidence from Nigeria**
  with P. Castaing, I. Contreras, A. Palacios-Lopez, A. Sagesaka, and P. Wollburg [Working Paper]
- **Impacts of Survey Design on the Measurement of Health Outcomes: Evidence from National Phone Surveys in Africa**
  with A. Ambel, T. Kilic, and P. Wollburg [Abstract]
- **The anatomy of survey measurement error: Experimental evidence from Bangladesh**
  with P. Castaing, I. Contreras, A. Palacios-Lopez, and P. Wollburg [Abstract]
- **Optimal survey mode choice and the long-run evidence gap**
  with E. Anderson [Abstract]

Public finance and governance

- **The cost of bureaucratic fragmentation: Business tax evasion and revenue mobilization in a low-income country**
  with S. Dietrich and R. Vincent [Latest Draft] [IFS Working Paper]
- **Tax amnesties in low-income countries: Evidence from administrative and experimental data in Uganda**
  with S. Dietrich [Abstract]
- **Artificial intelligence and accountable governance: Experimental evidence from firms in Uganda**
  with S. Dietrich [Abstract]

Economic shocks and vulnerability

- **Chronic and Transient Food Insecurity: Evidence from High-Frequency Panel Data in Sub-Saharan Africa**
  with D. Jolliffe and P. Wollburg *Revise & resubmit, Proceedings of the National Academy of Sciences (PNAS)*
  [Abstract]
- **Forced displacement and vulnerability: Evidence from high-frequency microdata across 11 LMICs**
  with S. Boukaka, H. Mugera, and P. Wollburg [Abstract]

Research Grants

- **J-PAL Governance Initiative** Principal Investigator (USD 50,000) 2024
  For project *Tax compliance and accountable governance*
- **Dutch Research Council (NWO)** Co-Principal Investigator (EUR 50,000) 2024
  For project *Tax compliance and accountable governance*
- **International Growth Centre (IGC)** Co-Principal Investigator (GBP 34,000) 2024
  For project *Tax compliance and accountable governance*
- **STEG (CEPR)** Principal Investigator (GBP 15,000) 2024
  For project *The effects of a tax amnesty on tax morale and tax climate*
- **World Bank Research Support Budget** Team member (EUR 29,500) 2024
  For project *Survey mode effects in development economics research*
- **World Bank Research Support Budget** Team member (EUR 27,500) 2022
  For project *Understanding and estimating COVID-19 vaccination attitudes, uptake, and barriers in Sub-Saharan Africa*

Honors and Fellowships

- **UNU-MERIT PhD Scholarship** Full cost of tuition and living expenses 2021–2025
- **Young Scholar** Global Labor Organization (GLO) 2023–2024
- **Marie Skłodowska-Curie Actions grant** European Union 2023
- **Vice Presidential Unit Award** World Bank, Development Economics Vice Presidency 2021
- **Carlo-Schmid Fellowship** German Academic Exchange Service (DAAD) 2019
- **Luca D'Agliano Prize** Best MSc dissertation, University of Oxford 2019
- **Deutschlandstipendium** University of Tübingen and German Central Bank 2015

Teaching Experience

- **Development Economics (Lecturer)** Undergraduate, ETH Zurich 2026
- **Policy Impact Evaluation (Tutor)** Master's, ETH Zurich 2026
- **Policy & Program Analysis (Co-organizer & TA)** PhD, UNU-MERIT. Student Eval.: 9.6/10 2022–2025
- **Public Policy Analysis (Lecturer)** Master's, UNU-MERIT 2022–2025
- **Public Policy Analysis (Tutor)** Master's, UNU-MERIT. Student Evaluation: 9.4/10 2022
- **Data Analysis with Stata** PhD, UNU-MERIT 2023–2024

Professional Service

Refereeing: Journal of Development Economics, Journal of Population Economics, Food Policy, Journal of Development Studies, European Journal of Development Research, BMJ Global Health, Vaccines, Global Public Health, Global Health: Science & Practice, Discover Social Science and Health, Social Sciences & Humanities Open

Organization: Dutch Development Economics PhD Workshop (2023, 2024), UNU-MERIT External Seminar Series (2022–2023), Dutch Development Economics Workshop (2022), UNU-MERIT Internal Conference (2022)

Languages and Technical Skills

Languages: German (native), English (fluent), Portuguese (working proficiency), Italian (basic)
Software: Stata (advanced), R (intermediate)